Operating Systems Assignment 1

Character Device Driver Implementation

---

1. Problem Statement

The objective of this assignment is to design and implement a Linux Character Device Driver that enforces ordered execution of operations within a specified time limit.

The driver must:

- Accept two parameters: `kernel_version` and `timer`
- Validate kernel version at insertion time
- Register a character device and expose it to user space
- Enforce the order: **read → write**
- Ensure operations are completed within a given time
- Use kernel wait queues for synchronization
- Display success or failure message upon module removal

---

2. Methodology

The implementation is divided into the following stages:

2.1 Module Parameter Handling

- `kernel_version` is passed as an integer array (major, minor)
- `timer` defines the time constraint in seconds

2.2 Kernel Version Validation

- The module extracts the running kernel version using `LINUX_VERSION_CODE`
- Compares it with the user-provided version
- Module insertion fails if mismatch occurs

2.3 Character Device Creation

- Device number allocated using `alloc_chrdev_region`

- Character device initialized using `cdev_init`
- Device added using `cdev_add`
- Device file created using `class_create` and `device_create`

2.4 Wait Queue Mechanism

- A wait queue is used to enforce ordering
- If write occurs before read, the process blocks
- It resumes only when read operation occurs

2.5 Timing Mechanism

- Global timer starts at module insertion using `jiffies`
- Timeout is also used in write operation when waiting for read
- Final validation is done during module removal

---

3. Explanation of Implementation

3.1 Module Initialization (`my_init`)

- Initializes wait queue
- Validates input parameters
- Checks kernel version compatibility
- Allocates device number and registers device
- Stores start time using `jiffies`

```
akshay@fedora:~/Desktop/ANS_B240142CS$ make
make -C /lib/modules/6.19.9-200.fc43.x86_64/build M=/home/akshay/Desktop/ANS_B240142CS modules
make[1]: Entering directory '/usr/src/kernels/6.19.9-200.fc43.x86_64'
make[2]: Entering directory '/home/akshay/Desktop/ANS_B240142CS'
  CC [M]  mymodule.o
  MODPOST Module.symvers
  CC [M]  mymodule.mod.o
  CC [M]  .module-common.o
  LD [M]  mymodule.ko
  BTF [M] mymodule.ko
Skipping BTF generation for mymodule.ko due to unavailability of vmlinux
make[2]: Leaving directory '/home/akshay/Desktop/ANS_B240142CS'
make[1]: Leaving directory '/usr/src/kernels/6.19.9-200.fc43.x86_64'
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo insmod mymodule.ko kernel_version=6,19 timer=50
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo journalctl -k | tail -5
Apr 07 17:29:19 fedora kernel: kernel_version_count = 2
Apr 07 17:29:19 fedora kernel: kernel_version = 6.19
Apr 07 17:29:19 fedora kernel: mymodule: version matched
Apr 07 17:29:19 fedora kernel: timer = 50 seconds
Apr 07 17:29:19 fedora kernel: major = 511, minor = 0
akshay@fedora:~/Desktop/ANS_B240142CS$
```

3.2 Read Operation (`my_read`)

- Sets `read_done = 1`
- Wakes up any blocked write operation
- Sends message to user: "Hello from kernel device"

```
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo cat /dev/mymodule
Hello from kernel device
akshay@fedora:~/Desktop/ANS_B240142CS$
```

3.3 Write Operation (`my_write`)

- If read has not occurred:
    - Process waits using wait queue
    - Timeout occurs after specified duration

- If read is done:
    - Username is copied from user space
    - Order is marked correct

```
akshay@fedora:~/Desktop/ANS_B240142CS$ echo "ans_christy" | sudo tee /dev/mymodule
ans_christy
akshay@fedora:~/Desktop/ANS_B240142CS$
```

3.4 Module Exit (`my_exit`)

- Calculates total elapsed time
- Checks:
    - Read completed
    - Write completed
    - Correct order
    - Time constraint satisfied
- Prints success or failure message

---

4. Execution and Results

4.1 Successful Execution

Steps:

1. Insert module with correct parameters
2. Perform read operation
3. Perform write operation within time
4. Remove module

Expected Result:

- No errors during execution
- Kernel log displays:

  `Successfully completed the actions within time`

```
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo insmod mymodule.ko kernel_version=6,19 timer=50
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo cat /dev/mymodule
Hello from kernel device
akshay@fedora:~/Desktop/ANS_B240142CS$ echo "ans_christy" | sudo tee /dev/mymodule
ans_christy
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo rmmod mymodule
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo journalctl -k | tail -10
Apr 07 17:37:16 fedora kernel: major = 511, minor = 0
Apr 07 17:37:21 fedora kernel: mymodule: read called
Apr 07 17:37:21 fedora kernel: mymodule: read called
Apr 07 17:37:26 fedora kernel: mymodule: write called
Apr 07 17:37:26 fedora kernel: mymodule: username stored = ans_christy
Apr 07 17:37:35 fedora kernel: mymodule: read_done = 1, username_written = 1, order_ok = 1
Apr 07 17:37:35 fedora kernel: mymodule: stored username = ans_christy
Apr 07 17:37:35 fedora kernel: mymodule: elapsed time = 18 seconds
Apr 07 17:37:35 fedora kernel: Successfully completed the actions within time. Username: ans_christy
Apr 07 17:37:35 fedora kernel: mymodule: removed
akshay@fedora:~/Desktop/ANS_B240142CS$
```

4.2 Failure Case 1: Write Before Read

Steps:

1. Insert module
2. Perform write before read

Result:

- Write operation blocks and times out
- Error message:

  `Connection timed out`

- Final result shows failure

```
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo insmod mymodule.ko kernel_version=6,19 timer=50
akshay@fedora:~/Desktop/ANS_B240142CS$ echo "ans_christy" | sudo tee /dev/mymodule
ans_christy
```

```
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo insmod mymodule.ko kernel_version=6,19 timer=50
akshay@fedora:~/Desktop/ANS_B240142CS$ echo "ans_christy" | sudo tee /dev/mymodule
ans_christy
tee: /dev/mymodule: Connection timed out
akshay@fedora:~/Desktop/ANS_B240142CS$
```

4.3 Failure Case 2: Time Limit Exceeded

Steps:

1. Insert module with small timer
2. Perform read
3. Delay beyond timer
4. Perform write

Result:

- Write succeeds but final validation fails
- Kernel log shows failure message

```
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo insmod mymodule.ko kernel_version=6,19 timer=5
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo journalctl -k | tail -5
Apr 07 17:43:09 fedora kernel: kernel_version_count = 2
Apr 07 17:43:09 fedora kernel: kernel_version = 6.19
Apr 07 17:43:09 fedora kernel: mymodule: version matched
Apr 07 17:43:09 fedora kernel: timer = 5 seconds
Apr 07 17:43:09 fedora kernel: major = 511, minor = 0
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo rmmod mymodule
akshay@fedora:~/Desktop/ANS_B240142CS$ sudo journalctl -k | tail -5
Apr 07 17:43:09 fedora kernel: major = 511, minor = 0
Apr 07 17:43:53 fedora kernel: mymodule: read_done = 0, username_written = 0, order_ok = 0
Apr 07 17:43:53 fedora kernel: mymodule: elapsed time = 43 seconds
Apr 07 17:43:53 fedora kernel: Failed to complete the actions correctly within time
Apr 07 17:43:53 fedora kernel: mymodule: removed
akshay@fedora:~/Desktop/ANS_B240142CS$
```

---

5. Conclusion

The implemented character device driver successfully enforces ordered execution using wait queues and validates time constraints using kernel timing mechanisms. The driver correctly handles both successful and failure scenarios, demonstrating synchronization and timing control in kernel space.

---

6. References

- https://embetronicx.com/linux-device-driver-tutorials/
- https://www.apriorit.com/dev-blog/195-simple-driver-for-linux-os

---